A COUNTEREXAMPLE TO THE ERDŐS–MAHLER CONJECTURE

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Abstract. We disprove a 1939 conjecture of Erdős and Mahler by constructing a Cantor set of irrational numbers ξ with $4 \le \mu(\xi) \le 11$, each having infinitely many regular continued-fraction convergents of the form $2^a/3^b$.

1. Introduction

For an irrational real number ξ, its *irrationality exponent* $\mu(\xi)$ is the supremum of the real numbers $\nu > 0$ for which $0 < |\xi - u/v| < v^{-\nu}$ holds for infinitely many reduced fractions u/v. Thus ξ is a Liouville number precisely when $\mu(\xi) = \infty$. Write p_n/q_n for the regular continued-fraction convergents of ξ, and $\mathrm{P}(m)$ for the largest prime factor of $|m|$, with $\mathrm{P}(\pm 1) = 1$.

In 1939, Erdős and Mahler [5, p. 18] proposed the following conjecture.

Conjecture (Erdős–Mahler)**.** If $\mathrm{P}(p_n q_n)$ is bounded for infinitely many n, then ξ is a Liouville number.

We disprove the conjecture even when the prime factors of the selected convergents are restricted to 2 and 3.

Theorem. *There exists a Cantor set $K \subset \mathbb{R}$ such that every $\xi \in K$ has infinitely many regular continued-fraction convergents of the form $2^a/3^b$, yet satisfies $4 \le \mu(\xi) \le 11$.*

The hypothesis of the Erdős–Mahler conjecture already implies transcendence by Ridout's theorem [11]; see also [2, Theorem 2.3, p. 29]. Our result shows that it does not force an infinite irrationality exponent.

Erdős and Mahler also proved that ξ is a Liouville number if $\mathrm{P}(q_{n-1} q_n q_{n+1})$ is bounded for infinitely many n [5, Theorem 2, p. 17]. Shorey [13] obtained effective refinements. The recurrence $q_{n+1} = a_{n+1} q_n + q_{n-1}$ links the three denominators in their theorem. In the conjecture, successive indices for which $\mathrm{P}(p_n q_n)$ is bounded may be arbitrarily far apart.

Fraenkel [6, 7] studied rational approximation with prescribed prime factors. Fraenkel and Borosh subsequently obtained non-Liouville numbers

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by allowing the numerator and denominator of the approximating fractions to have cofactors outside the prescribed sets of primes. When both cofactors are bounded, their Hausdorff dimension formula gives zero, and they explicitly left open whether non-Liouville numbers exist in this case [8, p. 459]. Bugeaud recorded the Erdős–Mahler conjecture as Problem 43 in his book [2, p. 215]. More recently, Bugeaud and Nguyen [3, Question 4.1, p. 32] restated it and constructed further examples of numbers with infinitely many smooth convergents.

Idea of the proof. We choose the successive convergents from the fractions $R(b) = 2^{\lceil b\alpha \rceil}/3^b$, where $\alpha = \log 3/\log 2$. To bound the irrationality exponent of the limit, we must control approximation at the intervening denominators. At each stage, the indices for which $R(b)$ lies in the prescribed interval form a set of returns of an irrational rotation. Its threefold sumset has cardinality comparable to that of the return set itself. Two triples of indices with the same sum give the same product of fractions $R(b)$. Within a suitable denominator range, sufficiently close rational approximations inherit these identities.

For small denominators, a congruence bounds the number of possible approximations. For large denominators, we use the inherited product identities. At high multiplicative rank, a sumset estimate gives a quadratic saving in the rank. At low rank, many quotients of triple products are rational powers. A uniform height bound from the p-adic Subspace Theorem allows us to choose the power exponent separately in each denominator range. Matveev's theorem then forces these quotients to equal the corresponding quotients of products of the $R(b)$, and rotation discrepancy bounds their number. Summing over the denominator ranges shows that a proportion tending to one of the returns is admissible.

The main theorem is deduced from a counting proposition in Section 2, whose proof occupies Section 3. The construction is ineffective owing to its reliance on the Subspace Theorem.

2. The counting proposition and the main theorem

Throughout, all logarithms are natural, and $\mathbb{N} = \{1, 2, \ldots\}$. Rational fractions have positive denominators. The notation $|I|$ denotes length when I is an interval and cardinality when I is a finite set. We write $F = O(G)$ or $F \ll G$ if $|F| \le CG$ for some constant $C > 0$, $F \asymp G$ if $F \ll G \ll F$, and $F = o(G)$ if $F/G \to 0$. Unless otherwise stated, all implicit constants are uniform in the varying parameters.

Set

$$\alpha = \frac{\log 3}{\log 2}, \qquad a(b) = \lceil b\alpha \rceil, \qquad R(b) = \frac{2^{a(b)}}{3^b} \quad (b \in \mathbb{N}).$$

Since α is irrational, $b\alpha < a(b) < b\alpha + 1$. Thus each $R(b)$ is a reduced rational in $(1, 2)$. For $t \in \mathbb{N}$, put

$$Q = 3^t, \qquad A = 18, \qquad N = Q^A, \qquad B = NQ^{-4},$$

and define

$$\zeta = R(t) + Q^{-4}, \qquad J = (\zeta, \zeta + Q^{-4}),$$
$$\mathcal{R} = \{b \in [Q, Q + N) \cap \mathbb{Z} : R(b) \in J\}.$$

Thus $|J| = Q^{-4}$ and $B = N|J| = Q^{14}$. Since $R(t) \le 2 - Q^{-1}$ and $2Q^{-4} < Q^{-1}$ for $Q \ge 3$, we have $J \subset (1, 2)$.

A reduced rational $u/v \in [1, 2]$ is *exceptional for* b if $u/v \ne R(b)$ and

$$\left|R(b) - \frac{u}{v}\right| < \frac{1}{16v^{11}}. \tag{1}$$

Write $\mathcal{R}_i$ for the returns $b \in \mathcal{R}$ admitting an exceptional fraction with $v > Q^3/6$, and put $\mathcal{R}_a = \mathcal{R} \setminus \mathcal{R}_i$. These returns are called *inadmissible* and *admissible*, respectively. Their dependence on t is suppressed except when needed.

Proposition 2.1 (Admissible returns)**.** *As $t \to \infty$ through the positive integers,*

$$|\mathcal{R}| \asymp B, \qquad |\mathcal{R}_i| = o(B), \qquad |\mathcal{R}_a| \asymp B.$$

In particular, $\mathcal{R}_a$ is nonempty for every sufficiently large t.

Before proving Proposition 2.1 in Section 3, we deduce the main theorem. We first construct a single counterexample, then use the abundance of admissible choices to obtain a Cantor set.

Proof of the theorem. By Proposition 2.1, choose an integer $t_0 \ge 4$ sufficiently large that $\mathcal{R}_a(t) \ne \varnothing$ for every $t \ge t_0$. Recursively select $t_{j+1} \in \mathcal{R}_a(t_j)$, and put

$$r_j = R(t_j), \qquad Q_j = 3^{t_j} \qquad (j \ge 0).$$

Since $t_{j+1} \in \mathcal{R}_a(t_j) \subseteq \mathcal{R}(t_j)$, the definition of $\mathcal{R}(t_j)$ gives

$$Q_j^{-4} < r_{j+1} - r_j < 2Q_j^{-4}, \qquad Q_{j+1} \ge 3^{Q_j} \ge 2Q_j. \tag{2}$$

Consider the closed intervals

$$K_j = [r_j + Q_j^{-4},\, r_j + 3Q_j^{-4}].$$

They lie in $(1, 2)$, since $r_j \le 2 - Q_j^{-1}$ and $3Q_j^{-4} < Q_j^{-1}$. Their left endpoints strictly increase, while (2) gives

$$r_{j+1} + 3Q_{j+1}^{-4} < r_j + \left(2 + \frac{3}{16}\right) Q_j^{-4} < r_j + 3Q_j^{-4}.$$

Hence $K_{j+1} \subset \operatorname{int} K_j$. Moreover, $Q_j \ge 2^j Q_0$, so $|K_j| \le 2Q_0^{-4}16^{-j}$. Their intersection therefore consists of a single point $\xi \in (1, 2)$, satisfying

$$Q_j^{-4} < \xi - r_j < 3Q_j^{-4} \qquad (j \ge 0). \tag{3}$$

If $\xi = u/v$ were rational, choose j with $Q_j^3 > 3v$. Since $\xi \neq r_j$, rational separation would give

$$|\xi - r_j| \geq \frac{1}{vQ_j} > 3Q_j^{-4},$$

contrary to (3). Thus ξ is irrational. For every j, $Q_j \geq 3$ also gives

$$|\xi - r_j| < \frac{3}{Q_j^4} < \frac{1}{2Q_j^2}.$$

Legendre's criterion [2, Theorem 1.8, p. 10] therefore shows that every r_j is a regular continued-fraction convergent of ξ. Their denominators are strictly increasing powers of 3, and their numerators are powers of 2. For each $0 < \nu < 4$, (3) gives $0 < |\xi - r_j| < Q_j^{-\nu}$ whenever $Q_j^{4-\nu} \geq 3$. Since $Q_j \geq 2^j Q_0$, this holds for infinitely many j, proving $\mu(\xi) \geq 4$.

For the upper bound, let $u/v \in [1, 2]$ be reduced, with $v \geq Q_0$, and choose j such that

$$Q := Q_j \leq v < Q_{j+1} =: S.$$

Since $t_j \geq 4$, we have $3^{t_j} > 12t_j$, and hence $Q^3 < 3^{Q/4} \leq S^{1/4}$. We distinguish three ranges.

(i) $Q \leq v \leq Q^3/6$. If $u/v = r_j$, then $v = Q$ and $|\xi - u/v| > v^{-4}$. Otherwise, rational separation and (3) give

$$|\xi - u/v| \geq \frac{1}{vQ} - \frac{3}{Q^4} \geq \frac{1}{2vQ} \geq \frac{1}{2v^2},$$

because $6v \leq Q^3$.

(ii) $Q^3/6 < v < S^{1/4}$. The reduced denominator of r_{j+1} is $S > v$, so $u/v \neq r_{j+1}$. Since $t_{j+1} \in \mathcal{R}_a(t_j)$, admissibility gives

$$|r_{j+1} - u/v| \geq \frac{1}{16v^{11}}.$$

As $S > v^4$ and $v \geq 3$,

$$\frac{3}{S^4} < \frac{3}{v^{16}} \leq \frac{1}{32v^{11}},$$

where the last inequality follows from $v^5 \geq 3^5 > 96$. Applying (3) at $j + 1$, we obtain

$$|\xi - u/v| \geq \frac{1}{16v^{11}} - \frac{3}{S^4} \geq \frac{1}{32v^{11}}.$$

(iii) $S^{1/4} \leq v < S$. Again $u/v \neq r_{j+1}$. Since $S \geq 3$ and $v < S$, we have $S^3 \geq 9S > 6v$. Rational separation yields

$$|\xi - u/v| \geq \frac{1}{Sv} - \frac{3}{S^4} \geq \frac{1}{2Sv} \geq \frac{1}{2v^5},$$

using $S \leq v^4$.

These estimates show that

$$|\xi - u/v| \geq \frac{1}{32v^{11}} \qquad (u/v \in [1, 2],\ v \geq Q_0).$$

For $1 \le v < Q_0$, the reduced fraction u/v differs from r_0, and $6v < Q_0^3$ gives

$$|\xi - u/v| \ge \frac{1}{vQ_0} - \frac{3}{Q_0^4} \ge \frac{1}{2vQ_0}.$$

Also, $1 + Q_0^{-1} \le r_0 \le 2 - Q_0^{-1}$, so

$$K_0 \subset \left[1 + \frac{1}{2Q_0},\, 2 - \frac{1}{2Q_0}\right].$$

Rationals outside $[1, 2]$ consequently have distance at least $1/(2Q_0)$ from ξ. Since $Q_0 \ge 81$, these bounds imply

$$\left|\xi - \frac{u}{v}\right| \ge \frac{1}{2Q_0 v^{11}}$$

for every reduced rational u/v with $v \ge 1$. In particular, $4 \le \mu(\xi) \le 11$.

A Cantor set of counterexamples. The preceding argument applies to every sequence of admissible indices. For $t \in \mathbb{N}$, put

$$K(t) = [R(t) + 3^{-4t},\, R(t) + 3^{1-4t}].$$

As shown above, $K(b) \subset \operatorname{int} K(t)$ whenever $b \in \mathcal{R}_a(t)$.

We claim that every sufficiently large t admits two admissible indices $b < c$ for which $K(b)$ and $K(c)$ are disjoint. Indeed, $|\mathcal{R}_a(t)| \asymp Q^{14}$, while $[Q, Q+N)$ can be covered by $O(\log Q)$ dyadic intervals. Thus two admissible indices satisfy $b < c < 2b$. Since $R(b)$ and $R(c)$ are distinct reduced fractions with denominators 3^b and 3^c,

$$|R(b) - R(c)| \ge 3^{-c} > 3^{-2b} > 2 \cdot 3^{1-4b}.$$

As $K(b) \subset [R(b), R(b) + 3^{1-4b}]$ and likewise for $K(c)$, these intervals are disjoint.

Starting from a sufficiently large t_0, iterate this selection. At level m we obtain 2^m pairwise disjoint closed intervals, each containing two intervals at the next level. Their lengths are bounded by $2Q_0^{-4}16^{-m}$, where $Q_0 = 3^{t_0}$. The intersection of the resulting nested unions is therefore a Cantor set K. Every $\xi \in K$ arises from an infinite sequence of admissible indices and hence satisfies the conclusions established above. □

3. Proof of Proposition 2.1

The proof rests on a contrast between the size of the return set $\mathcal{R}$ and that of its threefold sumset. We shall show that $|\mathcal{R}| \asymp B$, whereas $|3\mathcal{R}| \ll B$. Thus many distinct triples of return indices have the same sum. The special arithmetic form of $R(b)$ turns these additive relations into exact multiplicative identities, which persist under sufficiently close rational approximation.

Consequently, exceptional approximations in a fixed denominator range give rise to a set of rational numbers whose threefold product set has cardinality $O(B)$. This restriction is the basis of the counting argument: large multiplicative rank forces the exceptional set to be small, while small rank will be treated through rational power classes. We first establish the additive

and multiplicative relations, then exclude exceptional approximations in the remaining denominator ranges.

3.1. **Many returns, few sums.** For $x \in \mathbb{R}$, write $\{x\} = x - \lfloor x \rfloor$ and

$$\|x\| = \min_{m \in \mathbb{Z}} |x - m|.$$

Thus $\|x\|$ denotes distance to the nearest integer, not a vector norm. For a subset E of an additive group and an integer $k \geq 1$, let

$$kE = \{e_1 + \cdots + e_k : e_i \in E\},$$

where repetitions are allowed.

The set $\mathcal{R}$ is described by an irrational rotation. Put

$$\Phi(y) = 1 - \frac{\log y}{\log 2}, \qquad I = \Phi(J), \qquad \rho = |I|.$$

Since $\Phi(R(b)) = \{b\alpha\}$, the set $\mathcal{R}$ consists precisely of the indices $b \in [Q, Q + N) \cap \mathbb{Z}$ for which $\{b\alpha\} \in I$.

We use the estimate of Wu and Wang [14, Theorem 1]: for every $\varepsilon > 0$, there is an effective constant $\mathfrak{H}_\varepsilon$ such that

$$|p + q_1 \log 2 + q_2 \log 3| \geq \mathfrak{H}^{-4.1163051-\varepsilon}, \qquad \mathfrak{H} = \max\{|q_1|, |q_2|\} \geq \mathfrak{H}_\varepsilon, \quad (4)$$

for integers p, q_1, q_2. Taking $\varepsilon = 10^{-4}$, $p = 0$, $q_2 = -n$, and q_1 nearest to $n\alpha$ in (4) gives

$$\log 2 \, \|n\alpha\| \geq (2n)^{-4.1164051}$$

for sufficiently large positive integers n. By symmetry and the irrationality of α, there is $c_\kappa > 0$ such that

$$\|n\alpha\| \geq c_\kappa |n|^{-\kappa} \quad (n \in \mathbb{Z} \setminus \{0\}), \qquad \kappa = \frac{4117}{1000}, \quad \beta = 1 - \frac{1}{\kappa} = \frac{3117}{4117}. \quad (5)$$

Proposition 3.1 (Discrepancy)**.** *Uniformly for integers u, positive integers T, and circular intervals $\mathcal{I} \subset \mathbb{R}/\mathbb{Z}$,*

$$\#\{u \leq n < u + T : \{n\alpha\} \in \mathcal{I}\} = T|\mathcal{I}| + O(T^\beta). \quad (6)$$

Proof. Let $\widetilde{p}_i/\widetilde{q}_i$ denote the convergents of α. The standard convergent estimate [2, Corollary 1.4, p. 10] and (5) give $\widetilde{q}_{i+1} \ll \widetilde{q}_i^{\kappa}$.

On any $\widetilde{q}_i$ consecutive indices, the fractional parts of $n\alpha$ lie within $1/\widetilde{q}_i$ of a translate of the uniform grid of denominator $\widetilde{q}_i$. At most $O(1)$ grid points lie sufficiently close to the endpoints of $\mathcal{I}$ to affect the count. Hence each such interval of indices has discrepancy $O(1)$, uniformly in its initial index.

Decompose T greedily into convergent-denominator lengths. The number of blocks of length $\widetilde{q}_i$ is at most

$$\min\{C\widetilde{q}_i^{\kappa-1}, T/\widetilde{q}_i\}.$$

Use the first bound for $\widetilde{q}_i \leq T^{1/\kappa}$ and the second for larger denominators. Since $\widetilde{q}_{i+2} \geq 2\widetilde{q}_i$, summation gives $O(T^{1-1/\kappa}) = O(T^\beta)$. Endpoints and intervals crossing zero contribute only $O(1)$. □

The return set is large, but its threefold sumset remains of comparable cardinality.

Lemma 3.2 (Returns and their sums)**.** *For all sufficiently large t, there is a fixed constant $C_3 > 0$ such that*

$$|\mathcal{R}| \asymp B, \qquad |3\mathcal{R}| \le C_3 B. \tag{7}$$

Proof. Since

$$\frac{1}{2\log 2} \le |\Phi'(y)| \le \frac{1}{\log 2} \qquad (1 \le y \le 2),$$

we have

$$\frac{Q^{-4}}{2\log 2} \le \rho \le \frac{Q^{-4}}{\log 2}.$$

By (6), $|\mathcal{R}| = N\rho + O(N^\beta)$.

Every $\ell \in 3\mathcal{R}$ lies in $[3Q, 3Q+3N)$ and satisfies $\{\ell\alpha\} \in 3I \pmod 1$. Since $3\rho < 1$ for sufficiently large t, another application of (6) gives

$$|3\mathcal{R}| \le 9N\rho + O(N^\beta).$$

Finally,

$$\frac{N^\beta}{B} = Q^{4-A/\kappa} \longrightarrow 0,$$

because $A = 18 > 4\kappa$. Since $N\rho \asymp B$, both assertions follow. □

For shorter arcs, the discrepancy error may exceed the expected number of visits. We shall also need an estimate that retains the arc length.

Lemma 3.3 (Short-arc estimate)**.** *For a circular interval $\mathcal{I}$ of length $0 < \lambda \le 1$, any integer u, and any positive integer T,*

$$\#\{u \le n < u+T : \{n\alpha\} \in \mathcal{I}\} \ll T\lambda + (T^\kappa\lambda)^{1/(\kappa+1)} + 1. \tag{8}$$

Proof. Suppose first that $\mathcal{I}$ does not cross zero. The number of visits is bounded by the number of points of the lattice

$$\Lambda = \left\{ \left(\frac{n}{T}, \frac{n\alpha - r}{\lambda} \right) : n, r \in \mathbb{Z} \right\}$$

in a translate of the closed unit square. Its covolume is $\Delta = (T\lambda)^{-1}$.

Let λ_1 be the Euclidean length of a shortest nonzero lattice vector. It is primitive and extends to a lattice basis. The parallel lattice lines are separated by Δ/λ_1, and their lattice points by λ_1. Thus the square contains

$$O\big((1 + \lambda_1/\Delta)(1 + \lambda_1^{-1})\big)$$

lattice points. If it contains more than one point, then $\lambda_1 \le \sqrt{2}$, and this bound is $O(1 + \Delta^{-1} + \lambda_1^{-1})$.

For a shortest vector with nonzero first coordinate, (5) gives

$$\lambda_1 \ge \max\left\{ \frac{|n|}{T}, \frac{c_\kappa}{\lambda |n|^\kappa} \right\} \ge \left(\frac{c_\kappa}{T^\kappa \lambda} \right)^{1/(\kappa+1)}.$$

For a vertical vector, $\lambda_1 \geq 1/\lambda$, hence $\lambda_1^{-1} \leq 1$. Substituting these bounds proves (8). An interval crossing zero is split into two intervals. □

3.2. **From additive relations to multiplicative identities.** We now use the arithmetic form of the fractions $R(b)$. If two threefold sums of return indices coincide, the corresponding products have the same denominator, while their proximity forces their numerators to coincide. The resulting identities will also hold for sufficiently close rational approximations.

For $X \subset \mathbb{Q}_{>0}$ and $k \geq 1$, write

$$X^{[k]} = \{x_1 \cdots x_k : x_i \in X\},$$

again allowing repetitions.

Lemma 3.4 (Products determined by index sums)**.** *For sufficiently large t, each $\ell \in 3\mathcal{R}$ determines a unique integer $h(\ell)$ such that*

$$Y(\ell) := \prod_{i=1}^{3} R(b_i) = \frac{2^{h(\ell)}}{3^{\ell}} \quad \text{whenever } \ell = b_1 + \cdots + b_3, \quad b_i \in \mathcal{R}. \tag{9}$$

All the numbers $h(\ell) - \alpha\ell$ belong to a common interval of length 3ρ contained in $(0, 3)$. Moreover, for $\ell \neq \ell'$,

$$|Y(\ell) - Y(\ell')| \geq c_\kappa \log 2\, (3N)^{-\kappa}. \tag{10}$$

Proof. Two products corresponding to representations of the same ℓ have quotient an integral power of 2, and both lie in $(\zeta^3, (\zeta + Q^{-4})^3)$. Since

$$\left(\frac{\zeta + Q^{-4}}{\zeta}\right)^3 < 2$$

for sufficiently large t, their quotient must equal one. Thus (9) defines $h(\ell)$ uniquely.

Since $a(b) - \alpha b = 1 - \{b\alpha\}$ and $\{b\alpha\} \in I$, the quantities $h(\ell) - \alpha\ell$ lie in a common interval of length 3ρ contained in $(0, 3)$. For distinct $\ell, \ell' \in 3\mathcal{R}$,

$$\begin{aligned} |\log Y(\ell) - \log Y(\ell')| &= \log 2\, |h(\ell) - h(\ell') - \alpha(\ell - \ell')| \\ &\geq \log 2\, \|(\ell - \ell')\alpha\| \\ &\geq c_\kappa \log 2\, (3N)^{-\kappa}. \end{aligned}$$

Both products exceed one, so their difference is at least the difference of their logarithms. This proves (10). □

We next transfer these identities to the rational approximations whose number must be estimated. Fix a nonempty set $\mathcal{A} \subseteq \mathcal{R}$ and a denominator interval $(V, H]$. For each $b \in \mathcal{A}$, choose a reduced rational $s_b \in [1, 2]$, with denominator v_b, satisfying

$$V < v_b \leq H, \qquad s_b \neq R(b), \qquad |R(b) - s_b| < (16 v_b^{11})^{-1}. \tag{11}$$

Put $X = \{s_b : b \in \mathcal{A}\}$ and $Z = X^{[3]}$.

Lemma 3.5 (Product correspondence)**.** *There exists a fixed constant $C_2 > 0$ such that, for sufficiently large t, the conditions*

$$V^{11} > C_2 N^{\kappa}, \qquad V^{11} > 2H^6 \tag{12}$$

make $b \mapsto s_b$ injective and induce a bijection

$$\psi : 3\mathcal{A} \longrightarrow Z, \qquad \psi(b_1 + \cdots + b_3) = s_{b_1} \cdots s_{b_3}. \tag{13}$$

In particular,

$$|\mathcal{A}| \le |Z| = |3\mathcal{A}| \le C_3 B. \tag{14}$$

If $z = U/W = \psi(\ell)$ is reduced, then

$$1 \le z, Y(\ell) \le 8, \qquad U \le 8H^3, \quad W \le H^3, \qquad 0 < |Y(\ell) - z| < V^{-11}. \tag{15}$$

Proof. If $s_b = s_{b'}$ for distinct $b, b' \in \mathcal{A}$, then (11) gives $|R(b) - R(b')| < (8V^{11})^{-1}$. Applying Φ, we obtain

$$\|(b - b')\alpha\| < (8 \log 2\, V^{11})^{-1},$$

contrary to (5), since $0 < |b - b'| < N$ and C_2 is sufficiently large. Thus $b \mapsto s_b$ is injective.

For any $b_1, \ldots, b_3 \in \mathcal{A}$, telescoping products of numbers in $[1, 2]$ gives

$$\left| \prod_{i=1}^{3} R(b_i) - \prod_{i=1}^{3} s_{b_i} \right| \le 4 \sum_{i=1}^{3} |R(b_i) - s_{b_i}| < \frac{3}{4} V^{-11} < V^{-11}.$$

Equal index sums give equal smooth products by Lemma 3.4. The corresponding rational products therefore differ by less than $2V^{-11}$. As their reduced denominators are at most H^3, rational separation and the second condition in (12) force them to coincide. Hence ψ is well defined and surjective.

Conversely, if two rational products coincide, their smooth products differ by less than $2V^{-11}$. For distinct index sums, this contradicts (10) provided

$$C_2 > \frac{2 \cdot 3^{\kappa}}{c_{\kappa} \log 2}.$$

Thus ψ is injective. Fixing two summands and applying (7) proves (14).

The bounds on z, U, W in (15) follow by multiplying three fractions in $[1, 2]$ with denominators at most H. Furthermore, rational separation gives

$$\frac{1}{3^b v_b} \le |R(b) - s_b| < \frac{1}{16 v_b^{11}}, \qquad 16 v_b^{10} < 3^b.$$

Consequently, if $\ell = b_1 + \cdots + b_3$,

$$W \le \prod_{i=1}^{3} v_{b_i} < 16^{-3/10} 3^{\ell/10} < 3^{\ell}.$$

Since $Y(\ell)$ has reduced denominator 3^{ℓ}, we have $z \ne Y(\ell)$. The telescoping estimate proves the remaining assertion of (15). □

The preceding lemma bounds the size of the threefold product set independently of the denominator height. We shall combine this bound with its multiplicative rank.

For a finite nonempty $X \subset \mathbb{Q}_{>0}$, the *affine multiplicative rank* of X is the rank of the group generated by $\{x/x_0 : x \in X\}$, for any fixed $x_0 \in X$. This does not depend on the choice of x_0. For a prime p, let $v_p(x)$ denote the p-adic valuation of a positive rational x.

Lemma 3.6 (Multiplicative rank bound)**.** *Let $X \subset [1,2] \cap \mathbb{Q}$ be finite and nonempty. Put $M = |X|$, $\mathsf{P} = |X^{[3]}|$, and let d be its affine multiplicative rank. Then $X^{[3]}$ has rank d and*

$$Md^2 \le 6\mathsf{P} \qquad (d \ge 1). \tag{16}$$

Proof. Fix $x_0 \in X$ and put

$$E = \{(v_p(x/x_0))_p : x \in X\}.$$

This set contains zero and spans a d-dimensional real vector space. By unique factorization, $X^{[3]}/x_0^3$ corresponds to $3E$, which has the same rank. For $d \ge 1$, the Matolcsi–Ruzsa sumset theorem [9, Theorem 1.5] gives

$$|3E| \ge \binom{d+2}{2}\left(M - \frac{2d}{3}\right) \ge \frac{Md^2}{6},$$

since $d \le M - 1$. □

3.3. **Excluding small denominators.** We first exclude exceptional approximations with denominators below $V_* = Q^{A/2}/t$. The position of J immediately to the right of $R(t)$ restricts the possible rational fractions to a small number of residue classes. Combined with the short-arc estimate, this gives a negligible contribution.

For $\nu > 0$ and $0 < Y_1 < Y_2 \le \infty$, let $E_\nu(Y_1, Y_2)$ denote the number of indices $b \in \mathcal{R}$ for which there exists a reduced rational $u/v \in [1,2]$ such that

$$Y_1 < v \le Y_2, \qquad u/v \ne R(b), \qquad |R(b) - u/v| < (16v^\nu)^{-1}.$$

Thus $|\mathcal{R}_i| = E_{11}(Q^3/6, \infty)$.

Lemma 3.7 (Localized fractions)**.** *For $V \ge Q^3/6$ and sufficiently large Q, at most $10V^2/Q^4$ reduced fractions $u/v \in [1,2]$ with $V < v \le 2V$ have a radius-$(16v^9)^{-1}$ neighborhood intersecting J.*

Proof. Write $p = 2^{a(t)}$, so that $R(t) = p/Q$ and $\gcd(p, Q) = 1$. By the definition of J, every such fraction satisfies

$$\frac{v}{Q^3} - \frac{Q}{16v^8} < \Delta := Qu - pv < \frac{2v}{Q^3} + \frac{Q}{16v^8}.$$

Since $V < v \le 2V$ and $V \ge Q^3/6$, these inequalities give, for sufficiently large Q,

$$0 < \Delta < \frac{5V}{Q^3}.$$

For each integer Δ in this range, the congruence

$$pv \equiv -\Delta \pmod{Q}$$

determines a unique residue class modulo Q. It contains at most $V/Q + 1 \leq 2V/Q$ integers $v \in (V, 2V]$. Once v and Δ are fixed, so is $u = (pv + \Delta)/Q$. The number of possible fractions is therefore at most

$$\frac{5V}{Q^3} \cdot \frac{2V}{Q} = \frac{10V^2}{Q^4}.$$

□

We apply this bound with approximation exponent 9, which also controls the exceptional set defined by exponent 11.

Proposition 3.8 (Small denominators)**.** *Put $V_* = Q^{A/2}/t$. For sufficiently large t,*

$$E_{11}(Q^3/6, V_*) \ll Bt^{-2}. \tag{17}$$

Proof. Fix $V \geq Q^3/6$. By Lemma 3.7, there are at most $O(Q^{-4}V^2)$ reduced fractions with denominators in $(V, 2V]$ whose radius-$(16v^9)^{-1}$ neighborhoods intersect J.

Under Φ, each such neighborhood determines an interval of length $O(V^{-9})$. Summing the short-arc estimate (8) over these intervals gives

$$E_9(V, 2V) \ll NQ^{-4}V^{-7} + N^{\kappa/(\kappa+1)}Q^{-4}V^{\sigma} + Q^{-4}V^2,$$
$$\sigma = 2 - \frac{9}{\kappa+1} = \frac{1234}{5117} > 0.$$

Sum over dyadic intervals from $Q^3/6$ to V_*, truncating the last interval if necessary. The first term decreases geometrically, while the other two increase. The sums of the three terms are

$$O(BQ^{-21}), \qquad O(BQ^{-\vartheta A}t^{-\sigma}), \qquad O(Bt^{-2}),$$

respectively, where

$$\vartheta = \frac{9/2 - \kappa}{\kappa + 1} = \frac{383}{5117} > 0.$$

Here we used $B = NQ^{-4}$ and

$$N^{-1/(\kappa+1)}V_*^{\sigma} = Q^{-\vartheta A}t^{-\sigma}.$$

Since $A = 18$ and $Q = 3^t$, the first two contributions are $O(Bt^{-2})$. As $E_{11} \leq E_9$, this proves (17). □

3.4. **Exceptional heights and rational powers.** It remains to control exceptional approximations with denominators exceeding V_*. The first step is to show that their logarithmic heights are $o(N)$, uniformly over the return set. This follows from the p-adic Subspace Theorem. We then exploit rational power classes among the threefold products and use Matveev's theorem to separate pairs whose quotients do not agree with those of the corresponding smooth products.

For $x \in \mathbb{Q}^\times$ and a prime p, write $|x|_p = p^{-v_p(x)}$, where $v_p(-x) = v_p(x)$. For a positive integer n, put $[n]_6 = 2^{v_2(n)}3^{v_3(n)}$; the notation $|\cdot|_\infty$ denotes ordinary absolute value.

We use the following two-dimensional form of the p-adic Subspace Theorem. Let $\mathcal{V}$ be a finite set of places of $\mathbb{Q}$ containing infinity, and for each $v \in \mathcal{V}$ let $L_{1,v}, L_{2,v}$ be independent rational linear forms. For fixed $c, \eta > 0$, the nonzero primitive integer vectors $\mathbf{x} = (x_1, x_2)$ satisfying

$$\prod_{v \in \mathcal{V}} |L_{1,v}(\mathbf{x}) L_{2,v}(\mathbf{x})|_v \le c \max\{|x_1|, |x_2|\}^{-\eta} \tag{18}$$

lie in finitely many proper rational subspaces of $\mathbb{Q}^2$; see [12, p. 246, (1.3)–(1.5)] and [1, Corollary 7.2.5]. The constant c is absorbed by decreasing η outside a bounded set.

Related Subspace-Theorem estimates for rational approximations to quotients of power sums appear in Corvaja and Zannier [4]. The following lemma is tailored to the family $R(b)$.

Lemma 3.9 (Fixed-exponent separation)**.** *For every $\varepsilon > 0$, there exists $b_0 = b_0(\varepsilon)$ such that, for all integers $b \ge b_0$ and reduced rationals $u/v \in [1, 2]$ distinct from $R(b)$,*

$$|R(b) - u/v| \ge 3^{-\varepsilon b} v^{-2}. \tag{19}$$

Proof. Write $P = 2^{a(b)}$, $S = 3^b$, and $e = |Pv - Su| \ge 1$. Put

$$\mathfrak{g} = \gcd(Pv, Su), \qquad \mathsf{U} = \frac{Pv}{\mathfrak{g}}, \qquad \mathsf{V} = \frac{Su}{\mathfrak{g}}.$$

The pair (U, V) is positive and primitive. Since $\gcd(u, v) = 1$, every prime divisor of $\mathfrak{g}$ belongs to $\{2, 3\}$. Consequently,

$$[\mathsf{UV}]_6 = \frac{PS[uv]_6}{\mathfrak{g}^2}, \qquad |(\mathsf{U} - \mathsf{V})\mathsf{V}|_\infty = \frac{eSu}{\mathfrak{g}^2}.$$

It follows that

$$\begin{aligned} \Pi(\mathsf{U}, \mathsf{V}) &:= |(\mathsf{U} - \mathsf{V})\mathsf{V}|_\infty |\mathsf{UV}|_2 |\mathsf{UV}|_3 \\ &= \frac{eu}{P[uv]_6} \le 2v^2 |P/S - u/v|. \end{aligned}$$

Suppose that (19) fails for arbitrarily large b. Rational separation gives $|P/S - u/v| \ge (Sv)^{-1}$, so any such failure requires $v < S$. Hence, with $\mathcal{H} = \max\{\mathsf{U}, \mathsf{V}\}$, we have $\mathcal{H} \le 2Sv < 2S^2$ and

$$\Pi(\mathsf{U}, \mathsf{V}) < 2S^{-\varepsilon} \le 2^{1+\varepsilon/2} \mathcal{H}^{-\varepsilon/2}. \tag{20}$$

Apply (18) to (20), taking $\mathcal{V} = \{\infty, 2, 3\}$, the forms $(X - Y, Y)$ at infinity, and (X, Y) at each of the places 2 and 3. The primitive pairs (U, V) then lie on finitely many rational lines, each containing at most one positive primitive pair.

On the other hand, along a sequence of failures with $b \to \infty$,

$$0 < |\mathsf{U}/\mathsf{V} - 1| = \frac{|P/S - u/v|}{u/v} < S^{-\varepsilon} v^{-2} \longrightarrow 0.$$

This is impossible for a finite set of positive primitive pairs with $\mathsf{U} \neq \mathsf{V}$. $\square$

For each t, let W_t be the maximum of $\log v$ over exceptional fractions associated with indices $b \in \mathcal{R}$, taking $W_t = 0$ if there are none. This maximum is finite: rational separation and (1) give $16v^{10} < 3^b$.

We claim that

$$W_t/N \longrightarrow 0 \qquad (t \to \infty). \tag{21}$$

Indeed, fix $\varepsilon > 0$. For sufficiently large t, all $b \in \mathcal{R}$ satisfy $b \geq Q \geq b_0(\varepsilon)$. Combining (19) with (1) gives $v^9 < 3^{\varepsilon b}/16$, and therefore

$$W_t \leq \frac{\varepsilon \log 3}{9}(Q+N).$$

Since $Q/N \to 0$, we obtain $\limsup_{t\to\infty} W_t/N \leq \varepsilon \log 3/9$. Letting $\varepsilon \downarrow 0$ proves (21).

We now partition the remaining denominator range. Since $16v^{10} < 3^b$ and $b < Q + N$, every exceptional denominator is smaller than $F = 3^{(Q+N)/10}$. With $\theta = 21/20$, define consecutive intervals $(V_j, H_j]$ by

$$V_0 = V_*, \qquad H_j = \min\{V_j^\theta, F\}, \qquad V_{j+1} = H_j, \tag{22}$$

ending when $H_j = F$. There are $O(\log N)$ such intervals, since their logarithmic endpoints increase by the factor θ before truncation.

Let $\mathcal{A}_j$ be the set of indices $b \in \mathcal{R}$ admitting an exceptional fraction with denominator in $(V_j, H_j]$. For each such index choose the fraction of least denominator, then least numerator. Write X_j for the selected fractions and put

$$Z_j = X_j^{[3]}, \qquad \mathsf{P}_j = |Z_j|,$$

with d_j denoting the affine multiplicative rank of X_j.

For every nonempty $\mathcal{A}_j$, the conditions of Lemma 3.5 hold uniformly:

$$\frac{V_j^{11}}{N^\kappa} \geq \frac{V_*^{11}}{N^\kappa} = \frac{Q^{(11/2-\kappa)A}}{t^{11}} \longrightarrow \infty, \qquad \frac{V_j^{11}}{H_j^6} \geq V_j^{47/10} \longrightarrow \infty.$$

Thus the correspondence gives a bijection $\psi_j : 3\mathcal{A}_j \to Z_j$ and $\mathsf{P}_j \leq C_3 B$.

We next count pairs of products whose quotients are rational powers. For an integer $q \geq 2$, put

$$\mathcal{C}_j(q) = \#\{(z, z') \in Z_j^2 : z'/z \in (\mathbb{Q}_{>0}^\times)^q\}.$$

A pair counted by $\mathcal{C}_j(q)$ is *proportional* if

$$\frac{z'}{z} = \frac{Y(\ell')}{Y(\ell)}, \qquad \ell = \psi_j^{-1}(z), \quad \ell' = \psi_j^{-1}(z').$$

Let $\mathcal{D}_j(q)$ count such proportional pairs. Both counts concern ordered pairs and include the diagonal.

Lemma 3.10 (Rational power classes)**.** *For every nonempty block and integer $q \geq 2$,*

$$\mathcal{C}_j(q) = \sum_{\omega} n_\omega^2, \qquad \frac{\mathsf{P}_j^2}{q^{d_j}} \leq \mathcal{C}_j(q) \leq \mathsf{P}_j^2, \tag{23}$$

where n_ω denotes the number of elements of Z_j in the class $\omega \in \mathbb{Q}_{>0}^\times/(\mathbb{Q}_{>0}^\times)^q$.

Proof. Fix $z_0 \in Z_j$. By Lemma 3.6, the group generated by the ratios z/z_0, $z \in Z_j$, has rank d_j. Its image modulo rational qth powers therefore has cardinality at most q^{d_j}. The identity in (23) counts ordered pairs belonging to the same class. Since $\sum_\omega n_\omega = \mathsf{P}_j$, the lower bound follows from Cauchy–Schwarz; the upper bound is immediate. □

The lower bound in (23) will be combined with an upper bound for proportional pairs. The essential step is to show that a nonproportional rational-power pair must have widely separated indices. For this we need a version of Matveev's theorem retaining the height of the rational base in its coefficient parameter.

For a positive rational $g = r/s$ in lowest terms, write $\mathrm{ht}(g) = \log\max\{r, s\}$.

Proposition 3.11 (Matveev)**.** *Let $g \in \mathbb{Q}_{>0} \setminus \{1\}$, $\tau = \mathrm{ht}(g)$, and $m, n, q \in \mathbb{Z}$. If $\mathcal{L} = m\log 2 + n\log 3 - q\log g \neq 0$, then*

$$\log|\mathcal{L}| > -C_0\tau(1 + \log\mathfrak{B}), \tag{24}$$

where $C_0 = 10^{12}$ and $\mathfrak{B} = \max\{1, |q|, \tau^{-1}\max(|m|\log 2, |n|\log 3)\}$.

Proof. Apply [10, p. 1218, (1.3), and p. 1219, Corollary 2.3, (2.6)] over $\mathbb{Q}$ to the bases $2, 3, g$, with height parameters $\eta_1 = \log 2$, $\eta_2 = \log 3$, and $\eta_3 = \tau$. These parameters satisfy the requirements of [10, (2.4)], since $\tau \geq \log 2 > 0.16$ and $|\log g| \leq \tau$. The coefficient parameter in (1.3) is precisely $\mathfrak{B}$. Since $C_1(3) \leq 2^{38}$ and $2^{38}\log 2\log 3 < 10^{12}$, the asserted bound follows. □

Lemma 3.12 (Separation of nonproportional pairs)**.** *Put $\gamma = (4C_0)^{-1}$. For all sufficiently large V, let $(V, H] = (V_j, H_j]$ be a denominator interval, and let $q \geq 2$ be an integer satisfying*

$$\frac{C_0(1 + \log q)}{q} < \frac{1}{4}. \tag{25}$$

Every nonproportional pair counted by $\mathcal{C}_j(q)$, with corresponding indices ℓ, ℓ', then satisfies

$$|\ell' - \ell| > (\log V)e^{\gamma q}. \tag{26}$$

Proof. Write $z'/z = g^q$, $\tau = \mathrm{ht}(g)$, $\delta = |\ell' - \ell|$, and $w = \log V$. Nonproportionality and the injectivity of ψ_j imply $g \neq 1$ and $\delta \geq 1$. Consider

$$\mathcal{L} = (h(\ell') - h(\ell))\log 2 - (\ell' - \ell)\log 3 - q\log g.$$

By nonproportionality, $\mathcal{L} \neq 0$. Moreover, (15) and $|(\log)'(y)| \leq 1$ for $y \geq 1$ give

$$0 < |\mathcal{L}| < 2V^{-11} = 2e^{-11w}.$$

Since $|h(\ell') - h(\ell)| < 2\delta + 3 \le 5\delta$ and $\tau \ge \log 2$, the bound (24) yields

$$\log|\mathcal{L}| > -C_0\tau\,(1 + \log\max\{q, 10\delta/\tau\})\,.$$

By (15), the reduced numerator and denominator of z'/z are at most $8H^6$. Since $H \le V^\theta$ and $\theta = 21/20$,

$$q\tau = \mathrm{ht}(z'/z) \le 3\log 2 + 6\log H \le 3\log 2 + \frac{63}{10}w \le 11w$$

for sufficiently large w.

For fixed q, δ, the function

$$f_\delta(y) = y\,(1 + \log\max\{q, 10\delta/y\})$$

is increasing for $y > 0$: on its two branches the derivatives are $\log(10\delta/y) \ge \log q$ and $1 + \log q$, respectively.

Suppose that $\delta \le we^{\gamma q}$, and put $\tau_0 = 11w/q$. Then $\tau \le \tau_0$ and

$$\max\{q, 10\delta/\tau_0\} \le qe^{\gamma q}.$$

Using the monotonicity of f_δ, we obtain

$$\begin{aligned}\log|\mathcal{L}| &> -C_0 f_\delta(\tau) \ge -C_0 f_\delta(\tau_0)\\ &\ge -11w\left(\frac{C_0(1+\log q)}{q} + C_0\gamma\right) > -\frac{11w}{2},\end{aligned}$$

where the last inequality follows from (25) and $C_0\gamma = 1/4$. This contradicts $|\mathcal{L}| < 2e^{-11w}$ for sufficiently large w, proving (26). □

3.5. **The rank dichotomy and completion.** We now choose the rational power exponent according to the height of each denominator interval. The restriction (21) ensures that the separation in Lemma 3.12 exceeds the entire range of possible index sums. Thus every rational-power pair is proportional, and the remaining count reduces to rotation discrepancy.

For each j with $\mathcal{A}_j \ne \varnothing$, put

$$w_j = \log H_j, \qquad x_j = \log\left(\frac{N}{w_j}\right), \qquad K = 16C_0, \quad K_1 = K{+}1, \quad q_j = \lceil Kx_j\rceil.$$

Let $\mathcal{J} = \{j : \mathcal{A}_j \ne \varnothing\}$ and, when $\mathcal{J} \ne \varnothing$, set $\chi = \min_{j\in\mathcal{J}} x_j$. For the empty set, put $\chi = +\infty$ and interpret all sums over $\mathcal{J}$ as zero.

Lemma 3.13 (Proportionality at the chosen exponent)**.** *As $t \to \infty$ through values for which $\mathcal{J} \ne \varnothing$, one has $\chi \to \infty$. For all sufficiently large t and every $j \in \mathcal{J}$,*

$$\mathcal{C}_j(q_j) = \mathcal{D}_j(q_j), \qquad q_j \le K_1 x_j, \qquad x_j < \log N.$$

Proof. For $j \in \mathcal{J}$, some exceptional fraction has denominator $v \in (V_j, H_j]$. Since $H_j \le V_j^\theta$,

$$w_j \le \theta\log V_j < \theta\log v \le \theta W_t = o(N).$$

By (21),

$$\chi \geq \log\left(\frac{N}{\theta W_t}\right) \longrightarrow \infty.$$

Thus $q_j \to \infty$ uniformly over $\mathcal{J}$, and (25) holds for every q_j when t is sufficiently large. Since $x_j \geq \chi \geq 1$, we have $q_j \leq K_1 x_j$; also $H_j \geq V_* > e$ gives $x_j < \log N$.

If a pair counted by $\mathcal{C}_j(q_j)$ were not proportional, Lemma 3.12 and $\gamma K = 4$ would give

$$\begin{aligned} |\ell' - \ell| &> (\log V_j) e^{\gamma q_j} \geq \frac{w_j}{\theta} e^{4x_j} \\ &= \frac{N}{\theta} e^{3x_j} \geq \frac{N}{\theta} e^{3\chi} > 3N. \end{aligned}$$

But $\ell, \ell' \in 3\mathcal{A}_j \subseteq [3Q, 3Q + 3N)$, a contradiction. Hence every counted pair is proportional. □

Proportionality imposes congruences on the exponents of 2 and 3, while the rational heights bound the difference of the corresponding indices. Together with discrepancy, these restrictions give the following estimate.

Lemma 3.14 (Localization of proportional pairs)**.** *For every integer $q \geq 2$ and every nonempty denominator interval $(V_j, H_j]$,*

$$\frac{\mathcal{D}_j(q)}{\mathsf{P}_j} \ll \frac{Q^{-4}\log(2H_j)}{q^2} + \left(1 + \frac{\log(2H_j)}{q}\right)^{\beta} + 1. \tag{27}$$

Proof. Let $z = U/W$ and $z' = U'/W'$ be a proportional pair counted by $\mathcal{C}_j(q)$, written in lowest terms, with corresponding indices ℓ, ℓ'. Then

$$\frac{U'W}{UW'} = \frac{Y(\ell')}{Y(\ell)} = 2^{h(\ell')-h(\ell)} 3^{-(\ell'-\ell)}.$$

If $\ell' > \ell$, taking 3-adic valuations shows that $3^{\ell'-\ell} \mid UW'$. Since $UW' \leq 8H_j^6$ by (15), and the roles of the fractions can be interchanged,

$$|\ell' - \ell| \leq 6\log_3 H_j + \log_3 8.$$

Fix $z = \psi_j(\ell_0)$ and write $h_0 = h(\ell_0)$. Because z'/z is a rational qth power and

$$\frac{z'}{z} = 2^{h(\ell')-h_0} 3^{-(\ell'-\ell_0)},$$

unique factorization gives

$$\ell' = \ell_0 + qn, \qquad h(\ell') = h_0 + qk \qquad (n, k \in \mathbb{Z}).$$

The preceding height bound restricts n to an interval of $O(1 + \log(2H_j)/q)$ consecutive integers.

By Lemma 3.4, all $h(\ell) - \alpha\ell$ belong to one real interval of length 3ρ. Hence

$$h(\ell') - \alpha\ell' = h_0 - \alpha\ell_0 + q(k - n\alpha)$$

restricts $k - n\alpha$ to a fixed interval of length $3\rho/q < 1$. Each n therefore determines at most one integer k, while $\{n\alpha\}$ belongs to a circular interval of length $3\rho/q$.

Applying Proposition 3.1, the number of possible n is

$$O\left(\frac{\rho \log(2H_j)}{q^2} + \left(1 + \frac{\log(2H_j)}{q}\right)^{\beta} + 1\right).$$

Each pair (n, k) determines at most one z' through the bijection ψ_j. Summing over the P_j choices of z and using $\rho \ll Q^{-4}$ proves (27). □

The lower bound for rational power pairs now meets the upper bound for proportional pairs. By Lemmas 3.10 and 3.13,

$$\frac{\mathsf{P}_j^2}{q_j^{d_j}} \le \mathcal{C}_j(q_j) = \mathcal{D}_j(q_j).$$

Combining this with (27), and using $\log(2H_j) \le 2w_j$ and $(1+u)^{\beta} \le 1 + u^{\beta}$ for $u \ge 0$, gives

$$\mathsf{P}_j \ll Q^{-4} w_j q_j^{d_j - 2} + w_j^{\beta} q_j^{d_j - \beta} + q_j^{d_j}. \tag{28}$$

For small multiplicative rank, (28) controls P_j directly. For large rank, the quadratic saving in (16) makes the contribution summable over the logarithmic grid. This explains the use of threefold products: twofold products yield only a linear rank saving, insufficient for this summation.

Lemma 3.15 (The logarithmic grid)**.** *Put $s = \log\theta > 0$. The numbers x_j, $j \in \mathcal{J}$, belong to an arithmetic grid of spacing s, apart from at most one point corresponding to the final truncated interval. If $\chi \ge 1$, then for every fixed $c > 0$,*

$$\sum_{j \in \mathcal{J}} e^{-c x_j} \ll_{c,\theta} e^{-c\chi}, \tag{29}$$

$$\sum_{j \in \mathcal{J}} x_j^{-4/3} \le 2\chi^{-4/3} + \frac{3}{s}\chi^{-1/3}, \tag{30}$$

$$\sum_{j \in \mathcal{J}} e^{x_j/4} \ll_{\theta} N^{1/4}. \tag{31}$$

Proof. Before the final truncated interval, $\log H_{j+1} = \theta \log H_j$, hence $x_{j+1} = x_j - s$. The final interval contributes at most one additional point.

The geometric series over grid points $x_j \ge \chi$ gives (29). For the powers $x^{-4/3}$, comparison with an integral gives

$$\sum_{\substack{x \text{ on the grid} \\ x \ge \chi}} x^{-4/3} \le \chi^{-4/3} + s^{-1} \int_{\chi}^{\infty} u^{-4/3}\, du.$$

Adding at most $\chi^{-4/3}$ for the final point proves (30). Finally, $x_j < \log N$ by Lemma 3.13. Summing the increasing exponential backwards from this upper endpoint gives (31). □

We can now divide the denominator intervals according to whether d_j is smaller or larger than $x_j^{2/3}$. In the first case, the factor $q_j^{d_j}$ grows sufficiently slowly for (28) to be effective. In the second, the quadratic rank bound gives a summable estimate.

Proposition 3.16 (Large denominators)**.** *If $\mathcal{J} \neq \varnothing$, then for all sufficiently large t,*

$$\frac{1}{B}\sum_{j\in\mathcal{J}} |\mathcal{A}_j| \ll \chi^{-1/3} + e^{-3\chi/4} + Q^{4-A/\kappa}e^{-(\beta-1/4)\chi} + Q^4 N^{-3/4} = o(1). \tag{32}$$

For $\mathcal{J} = \varnothing$, the sum is zero.

Proof. Since K_1 is fixed, for all sufficiently large x,

$$x^{2/3}\log(K_1 x) \leq x/4.$$

By Lemma 3.13, $\chi \to \infty$, so this holds for all x_j, $j \in \mathcal{J}$, when t is large.

Suppose first that $d_j \leq x_j^{2/3}$. Since $q_j \leq K_1 x_j$,

$$q_j^{d_j} \leq \exp\{x_j^{2/3}\log(K_1 x_j)\} \leq e^{x_j/4}.$$

Using (28), $q_j \geq 2$, and $|\mathcal{A}_j| \leq \mathsf{P}_j$, we obtain

$$\frac{|\mathcal{A}_j|}{B} \ll e^{-3x_j/4} + Q^{4-A/\kappa}e^{-(\beta-1/4)x_j} + B^{-1}e^{x_j/4},$$

because $w_j/N = e^{-x_j}$ and $Q^4 N^{\beta-1} = Q^{4-A/\kappa}$. Here $\beta - 1/4 = 8351/16468 > 0$. Summing over these intervals using (29) and (31) gives

$$\frac{1}{B}\sum_{\substack{j\in\mathcal{J}\\ d_j\leq x_j^{2/3}}} |\mathcal{A}_j| \ll e^{-3\chi/4} + Q^{4-A/\kappa}e^{-(\beta-1/4)\chi} + Q^4 N^{-3/4},$$

where $B = NQ^{-4}$ was used in the last term.

If $d_j > x_j^{2/3}$, the multiplicative rank bound (16) and Lemma 3.5 give

$$|\mathcal{A}_j| \leq \frac{6\mathsf{P}_j}{d_j^2} \leq \frac{6C_3 B}{d_j^2} < \frac{6C_3 B}{x_j^{4/3}}.$$

By (30), the sum over these intervals is $O(B\chi^{-1/3})$.

Combining the two cases proves (32). Every term on its right tends to zero: $\chi \to \infty$, $A = 18 > 4\kappa$, and

$$Q^4 N^{-3/4} = Q^{4-3A/4} \longrightarrow 0.$$

□

We have now excluded all but a negligible proportion of the returns admitting exceptional rational approximations.

Proof of Proposition 2.1. By Lemma 3.2, $|\mathcal{R}| \asymp B$. It remains to show that $|\mathcal{R}_i| = o(B)$.

Every exceptional fraction associated with $b \in \mathcal{R}$ satisfies $16v^{10} < 3^b$, hence $v < F$. The intervals treated in Proposition 3.8 and Proposition 3.16 therefore cover all inadmissible returns:

$$|\mathcal{R}_i| = E_{11}(Q^3/6, \infty) \le E_{11}(Q^3/6, V_*) + \sum_{j \in \mathcal{J}} |\mathcal{A}_j|.$$

When $\mathcal{J} \neq \varnothing$, the two propositions yield

$$\frac{|\mathcal{R}_i|}{B} \ll t^{-2} + \chi^{-1/3} + Q^{4-A/\kappa} + Q^4 N^{-3/4} \longrightarrow 0,$$

using $e^{-3\chi/4} \ll \chi^{-1/3}$ and $e^{-(\beta-1/4)\chi} \le 1$. When $\mathcal{J} = \varnothing$, the same conclusion follows directly from Proposition 3.8.

Thus $|\mathcal{R}_i| = o(B)$, and consequently

$$|\mathcal{R}_a| = |\mathcal{R}| - |\mathcal{R}_i| \asymp B.$$

This completes the proof of Proposition 2.1. □

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AI use declaration

The author used ChatGPT and Codex to review the exposition, check steps in the proofs, and compare cited statements with their sources. The author takes responsibility for the mathematical arguments and the accuracy of the references.

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